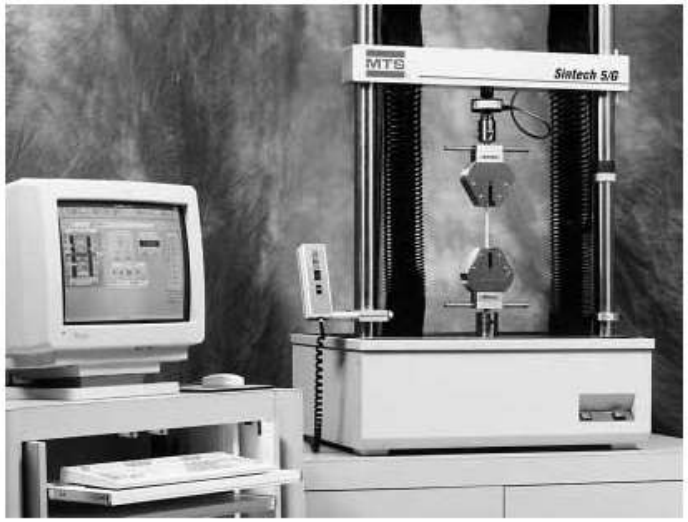
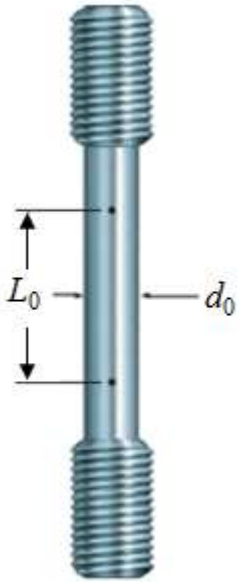
Chapter 3 – Stress-strain relation: (Constitutive equations)

3.1 Introduction

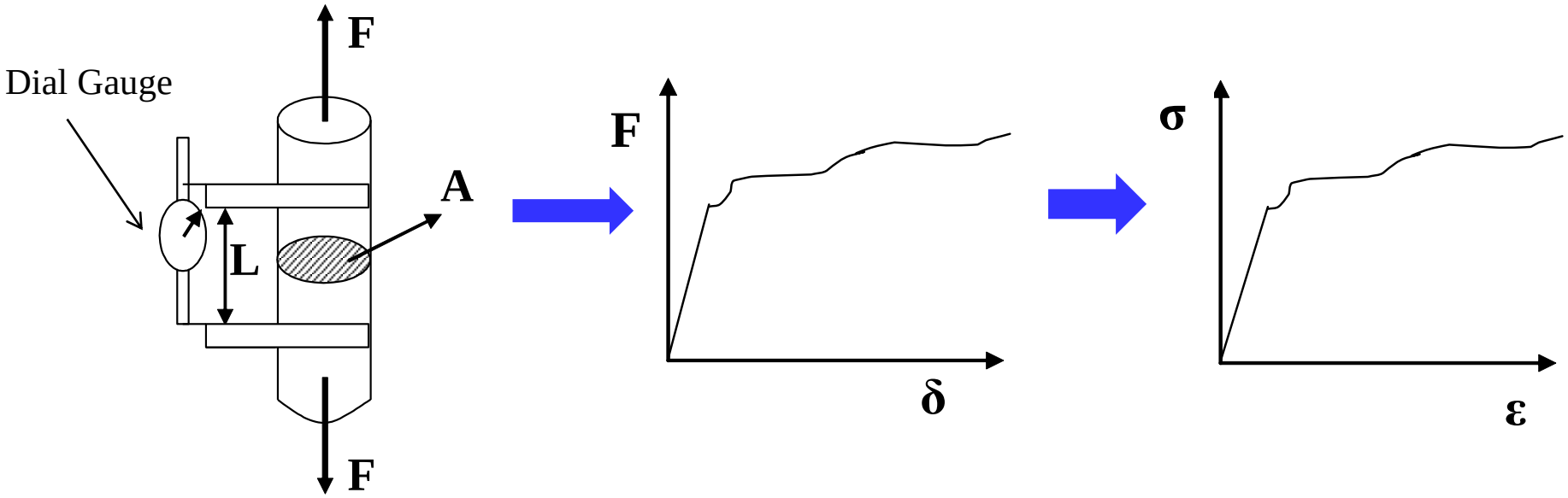
Stress-Strain relationship can be determined from the ***tension or compression test***.



Tensile-test machine with automatic data-processing system



In order to relate stress and strain, we carry out an experiment. In this experiment a cylindrical specimen is placed in a tensile testing machine and stretched. As the specimen stretches, the elongation and the load are measured.

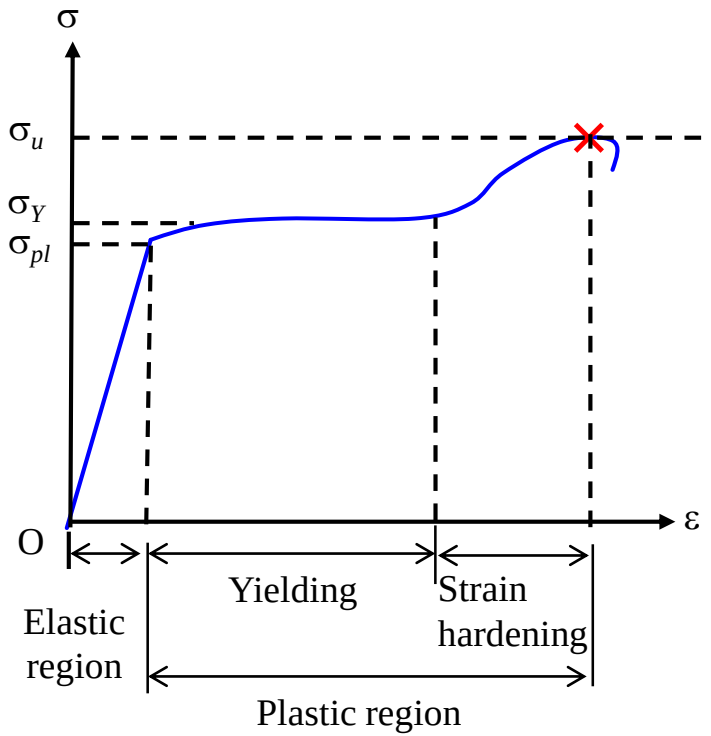


- *Nominal stress* or *engineering stress* is obtained by dividing the applied load *F* by the specimen's original cross-sectional area, $\sigma = F / A_0$.

- *Nominal strain* or *engineering strain* is obtained by dividing the change in the specimen's gauge length (elongation – δ) by the specimen's original gauge length, $\varepsilon = \delta / L_0$.

3.2 The normal stress – normal strain diagram

- Linearly elastic behaviour
  - Linearly elastic, stress is *proportional* to the strain.
  - Material is said to be linearly elastic.
- Yielding
  - Increase in stress above proportional limit will cause material to *deform permanently*.
- Strain hardening.
  - After yielding, a further load will reach, to the ultimate stress.
- Fracture/rupture
  - Specimen breaks at the ***ultimate stress*** or ***fracture stress***.

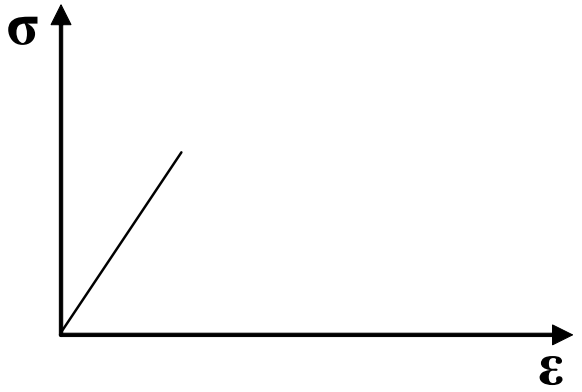


- σ_{pl}: Proportional limit
- σ_Y: Yield stress
- σ_u: Ultimate stress

For most materials, σ_{pl} and σ_Y are very close.

In this subject, unless we are told otherwise, we consider the linearly elastic materials. So, the yield stress and the proportional limit are same.

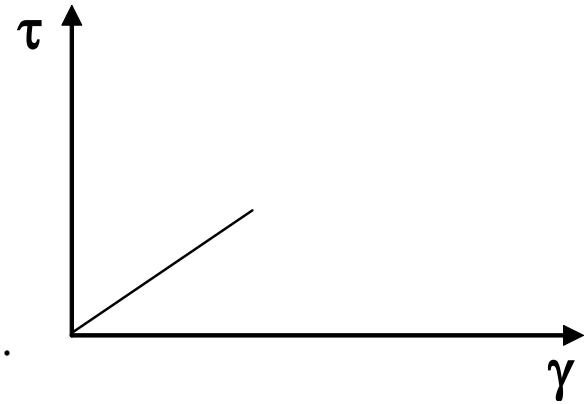
3.3 Linear elasticity, Hooke's law



Empirical/experimental observations for most solid materials **under small deformation** show that the stress and strain follow below law:

Hooke's Law

$$\sigma = E\varepsilon \qquad \tau = G\gamma \qquad (3.1)$$



E: is the Young's modulus or the modulus of elasticity;

G: is the shear modulus of elasticity.

Typical values of *E* are measured in GPa (10^9 Pa) or MPa (10^6 Pa).

For structural steel: $E = 200$ GPa , $\sigma_y = 250$ MPa , $\sigma_u = 400$ MPa

Typical values of *G* are measured in GPa (10^9 Pa) or MPa (10^6 Pa).

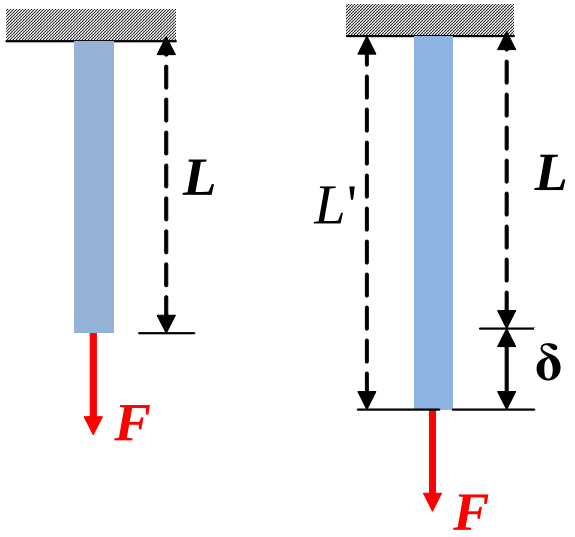
For structural steel: $G = 75$ GPa

Under small deformation and when the load is released, the stress obviously returns to zero. The strain reduces in accordance with Eq. (3.1). Therefore, if the body is loaded to less than the yield point, when the load is released the strain will return to zero – the body returns to its original shape and size.

How to find elongation δ based on Hooke's Law?

$$\sigma = \frac{F}{A} \qquad \sigma = E\varepsilon \qquad \varepsilon = \frac{\delta}{L}$$

$$\delta = \frac{FL}{EA} \qquad (3.2)$$



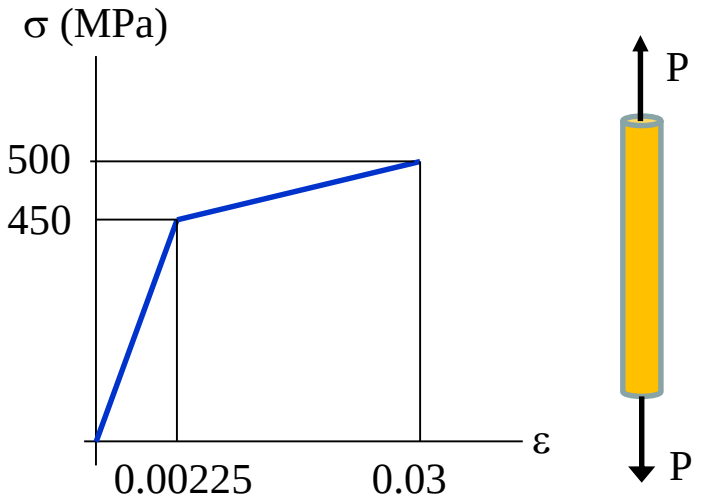
You should remember these equations and be able to apply them.

What do δ depend on?

- force *F*,
- cross-section area *A*,
- length *L*
- material.

Example 1: The material for the 50-mm-long specimen (with a diameter of 20 mm) has the stress-strain diagram shown below. If $P = 100$ kN, determine:

1. The Young's modulus of this specimen.
2. The elongation of this specimen



Solution:

$$E = \frac{450}{0.00225} = 200{,}000 \text{ MPa} = 200 \text{ GPa}$$

$$A = \pi(10)^2 = 314.159$$

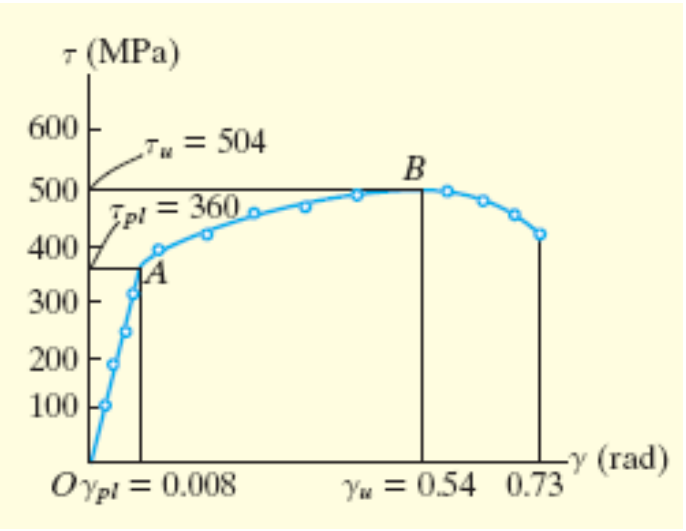
$$\sigma = \frac{P}{A} = \frac{100 \times 10^3}{314.159} = 318.31 \text{ MPa}$$

The stress is below the proportional limit.

$$\varepsilon = \frac{\sigma}{E} = \frac{318.31}{200{,}000} = 0.00159$$

$$\delta = \varepsilon L = 0.00159 \times 50 = 0.0796 \text{ mm}$$

Example 2: A specimen of titanium alloy is tested in torsion and the shear stress– strain diagram is shown. Determine the proportional limit, the shear modulus G and the ultimate shear stress.



Solutions

- By inspection, the graph ceases to be linear at point *A*. Thus, the proportional limit is

$$\tau_{pl} = 360\ \text{MPa}$$

- The shear stain corresponding to the proportional limit is

$$\gamma_{pl} = 0.008$$

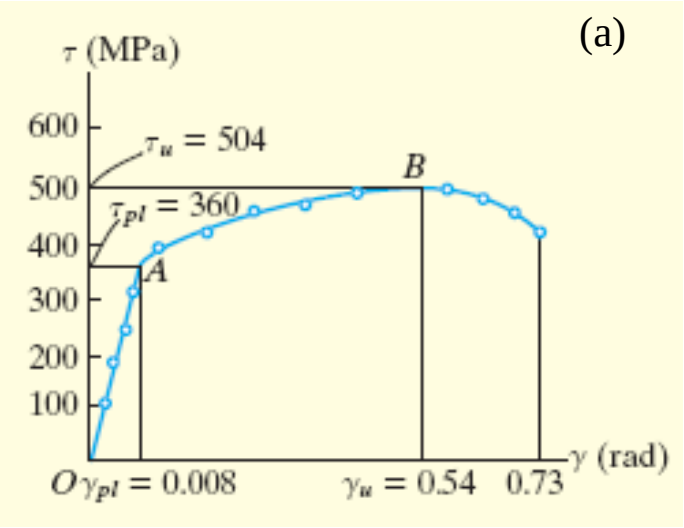
- Thus, the shear modulus is

$$G = \frac{\tau_{pl}}{\gamma_{pl}} = \frac{360}{0.008} = 45{,}000\ \text{MPa} = 45\ \text{GPa}$$

- The specimen breaks at B. The maximum shear stress is at point *B*. Thus the ultimate stress is

$$\tau_u = 504\ \text{MPa}$$

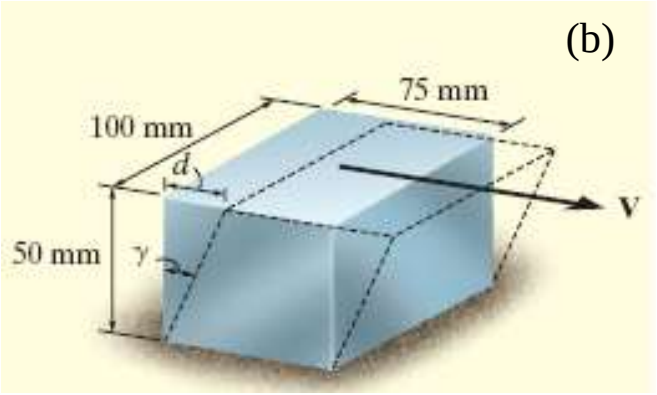
Example 3: For the same material of example 2 as shown in Fig. (a), determine the maximum distance *d* that the top of a block of this material, shown in Fig. (b), could be displaced horizontally if the material behaves elastically when acted upon by a shear force *V*. What is the magnitude of *V* necessary to cause this displacement?



Solutions

- Since the angle is small, the top of the block will be displaced horizontally by

$$\tan(0.008\ \text{rad}) \approx 0.008 = \frac{d}{50\ \text{mm}} \Rightarrow d = 0.4\ \text{mm}$$



- From the figures, the shear strain under this force *V* is 0.008, and the corresponding shear stress is 360 MPa.

- The shear force *V* needed to cause the displacement is

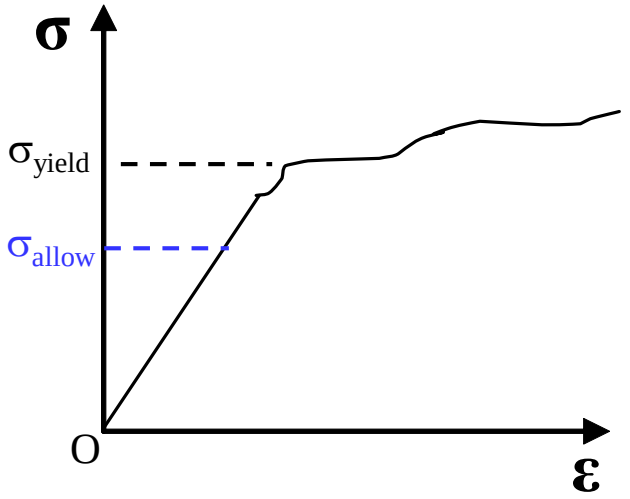
$$\tau_{avg} = \frac{V}{A}; \quad 360\ \text{MPa} = \frac{V}{75 \times 100} \Rightarrow V = 2{,}700\ \text{kN}$$

3.4 Factor of safety

When designing something, what is the maximum stress that we should allow it to have?

Yielding tensile strength would cause permanent deformation under normal loads. We therefore use yield strength.

However, if we allow the maximum expected loading, to cause a maximum stress equal to the yield stress, what happens if we unexpectedly go past the maximum expected loading ?



Therefore, when designing structure, the maximum stress that we should allow it to have must be less than the yield stress of the material.

We therefore define a quantity called Factor of Safety (F.S.):

$$\text{F.S.} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{allowable}}} > 1$$

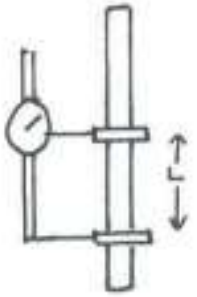
For example, if we specify a factor of safety of 3 to a material which has a yield stress of 200 MPa, the allowable stress you can apply to the material is

$$\sigma_{\text{allow}} = \frac{\sigma_{\text{yield}}}{\text{F.S.}} = \frac{200\ \text{MPa}}{3} = 66.7\ \text{MPa}$$

3.5 Measuring stress and strain



* For the simple axial loading of a bar, stress is constant, so we can measure F and A and calculate stress from equation $\sigma = \frac{F}{A}$

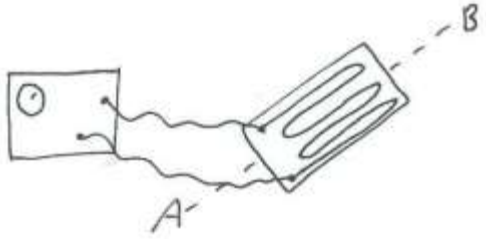


* The strain is also constant, so we can use a dial gauge to measure the elongation (δ), and calculate strain from equation $\varepsilon = \frac{\delta}{L}$

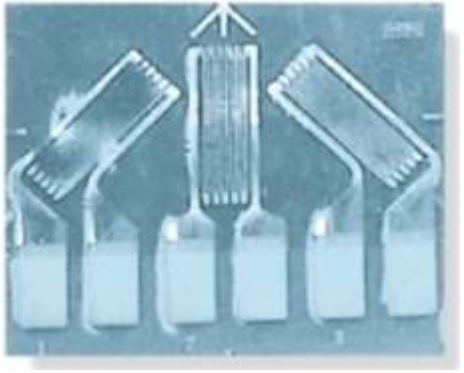


* In general, however, the stress and strain vary everywhere. We can measure the normal strain at any point on the surface by attaching a *strain gauge*.

* The electrical resistance of the gauge changes as it stretches in the direction AB. This allows us to calculate the normal strain in the direction AB. We can then use the Hooke's law ($\sigma = \varepsilon E$) to calculate the normal stress in the direction AB.



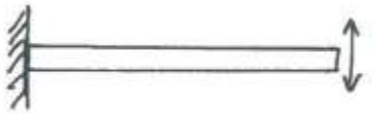
* A *strain gauge* measures normal strain in one direction only. To measure the strain in more than one direction we use several non-parallel strain gauges.



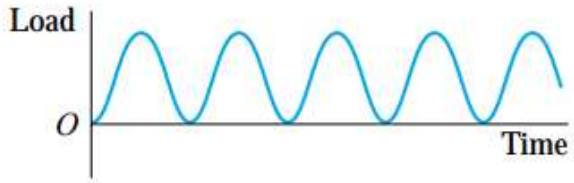
3.6 Fatigue

Does a tensile test tell us everything about whether a material will fail?

Is it possible for a component to fail at less than the yielding stress?



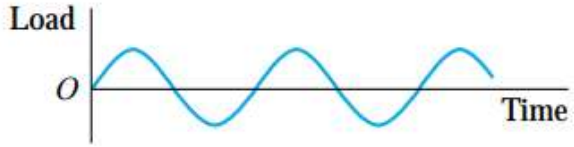
A tensile test applies a steady load to a component. In practical, however, components are often loaded and unloaded repeatedly.



(a)

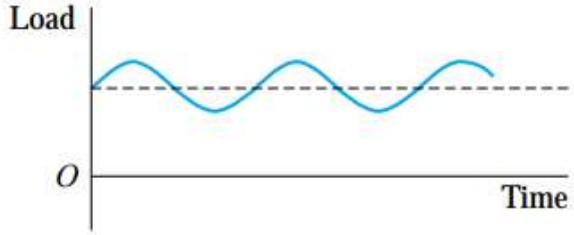
The top and bottom surfaces are repeatedly stretched and compressed. A small crack (microcrack) on the surface will open a bit more with each cycle, until the component fails completely.

A rotating shaft is an ideal place for fatigue. Shafts have to be designed to survive millions or even billions of cycles.



(b)

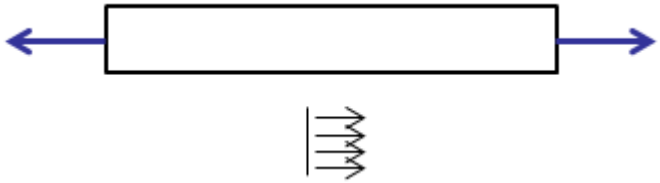
* Since fatigue failures usually begin with a microscopic crack at a point of high localized stress (that is, at a stress concentration), the condition of the surface of the material is extremely important. Highly polished specimens have higher endurance limits than rough surfaces.



(c)

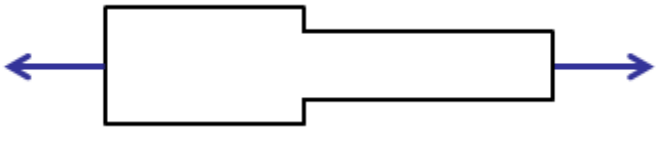
3.7 Stress concentrations

In a uniform cross-section, well away from the ends, the stress is uniform over the entire cross-section (Fig. (a)).

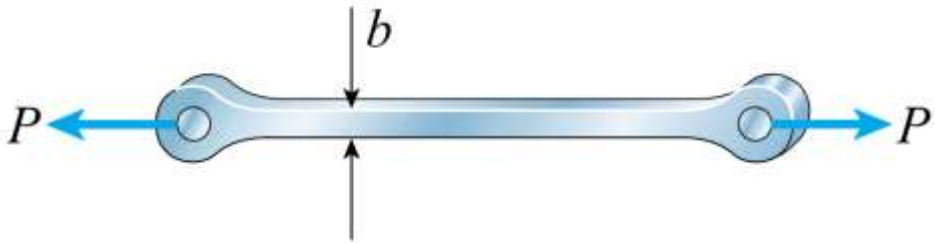


(a)

However, when a component has sharp corners (Fig. (b)), the stress will be concentrated near the corners. In the corners, the stress can be greater than it would be if the entire component were made from the thinner material. It is therefore best to avoid designing components with sharp corners. Fig. (c) shows a better design.



(b)



(c)

Quick review

Stress, Strain and Their relationship

Stress

Normal stress: $\sigma = \frac{F}{A}$ Shear stress: $\tau = \frac{V}{A}$

Strain

Normal strain: $\varepsilon = \frac{\delta}{L}$ Shear strain: $\gamma_{xy} = \text{Change of angle}$

Relationship between stress and strain – Hooke's law

$$\sigma = E\varepsilon$$

$$\tau = G\gamma$$

$$\delta = \frac{FL}{EA}$$